

LONDON FIRE BRIGADE PERSONNEL PLACEMENTS

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- My previous work
- Matching and Market Design
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INTRODUCTION

- Who am I?
 - Lecturer in Economics, Queen Mary University of London
 - PhD in economics, University of Oxford
- What do I want?
 - Research output: Publications, Grants, Policy reports
- What do I do?
 - I study allocation problems, personnel matching, and “market design”
 - Evaluating impacts of design changes, or assessing demand for certain policies
 - Previously worked on [Food banks](#), [Council Housing](#), Home Exchange, Ride Share, Doctor Allocation

INTRODUCTION TO MATCHING AND “MARKET DESIGN”

(WHERE I’M COMING FROM)

- What is a “market”? A place where matches happen
 - The “marriage market” – a place singles go to match with other singles
- Market Design is the study of how to organise centralised “exchanges”
 - Council houses, doctors, school places
 - Usually, markets we don’t want to leave to “market forces”
- A key focus on balancing efficiency and equity/fairness
 - Don’t want council houses to be vacant, or people to be stuck on endless waiting lists
 - Don’t want doctors stuck in jobs they dislike, wasting their skills

MY PREVIOUS WORK

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- Food banks [Feeding America] [DETAILS](#)
 - 100K tonnes of food distributed (pa) to their network of food banks. My PhD studied their “Choice System”.
 - Long distance ride sharing [Blablacar] [DETAILS](#)
 - Matching is mostly last minute, with ~50% unfilled seats. Trialling a mechanism to encourage earlier matches
 - Council Housing [interrupted by COVID] [DETAILS](#)
 - Working with 6 UK councils to improve the design of their waiting lists for council houses
 - How much do people want “any” house vs a “specific” house - vertical vs horizontal differences
 - Using recent advances in queuing theory (...yes...) we designed a policy to decrease waiting times by 40%
 - Doctor allocation [in progress] [DETAILS](#)
 - Junior doctors get placed in Foundation roles through an annual centralised matching procedure
 - In 2023 they changed the matching algorithm, requiring doctors to think strategically, and changed how doctors are ranked
 - Are doctors correctly strategising? Is anyone falling through the cracks? Impacts on doctor retention?



SCHOOL CHOICE:

- How do we allocate students to high schools?
 - Students want to be in good schools, and close to home
 - Schools have “preferences” over students (Siblings, Commuting Zones, Diversity, Grades?)
- In the UK this is centralised
 - Students rank schools, Schools “rank” students
 - An algorithm assigns students to schools
- How should we design this algorithm?

[DETAILS](#)

DOCTOR ALLOCATION:

FROM ROTH (1984)

- Since 1950s the job market for medical residents in the USA has been centralised
 - Doctors rank Hospitals, Hospitals rank Doctors, then an algorithm forms the match
- Matching is performed using an algorithm known as “Deferred-Acceptance”
 - Doctors have an incentive to rank hospitals truthfully – No Strategising
 - This ensures the resulting match is “stable” [No blocking pairs]
- However, this problem is much simpler than the one faced by the fire brigade
 - New firefighters “arrive” in batches
 - Firefighters join transfer lists to move around every few years

TEACHER ALLOCATION: THE PROBLEM

FROM COMBE, TERRIER, AND TERCIEUX (2022,2025, FORTHCOMING)

[OTHER PROBLEMS](#)

- Teachers have preferences over schools, schools want the best teachers
- This is a fundamentally dynamic problem:
 - New teachers move around to build experience, old teachers move somewhere cozy to settle
- In many countries, allocation is centralised and happens *each year*
 - Teachers rank schools – the default is to put their current school at the top
 - Schools rank teachers, and an [algorithm](#) assigns teachers to their new (or old) schools
- This algorithm could be adapted for to facilitate transfers across the LFB
 - It can be adapted to incorporate various constraints, as well as other priorities of the brigade

TEACHER ALLOCATION: DYNAMICS

FROM COMBE, TERRIER, AND TERCIEUX (2022,2025, FORTHCOMING)

[OTHER PROBLEMS](#)

- We want a good distribution of teachers across schools
 - Disadvantaged schools often struggle to attract good teachers / retain experienced teachers
 - We could just **pay** more at disadvantaged schools (this is undesirable)
- In several countries teachers get extra priority if they taught at a disadvantaged school
 - This gives them an extra incentive to go low SES
 - But, how to design this priority is important – you risk only having inexperienced teachers
- A similar policy is used to encourage Cadets in the U.S. military to enlist for longer terms
 - In return, they get higher priority at their preferred placements
- This type of policy could help the brigade incentivise firefighters to go to Command Unit
 - E.g. Spend six years at an Command Unit, then go straight to the top of the priority list

[DETAILS](#)



WHAT I CAN DO FOR YOU

- Identifying potential transfers:
 - Find pairs and chains of transfers based on lists --- Using insights from Teacher Allocation
 - Suggesting transfers, and identifying key blockages
- Skills incentives:
 - FRU training is costly, don't want people to get the training, then quickly transfer out
 - Don't want to put people off training if they think they'll get stuck at that station ["hold up"]
- Initial allocations:
 - Firefighters given initial allocation according to commute time and vacancies
 - New firefighters allocated in batches, use overlapping batches to improve matches (!!!)

MY GOALS

- I want to do research
 - Understand your specific challenges
 - Work with you to investigate remedies
 - Help me have “impact”
- I want to write and publish papers
 - Based on data sharing agreements (E.g. I need to see transfer lists)
- I want to apply for grants
 - Internal grants (~20k), external grants (~200k)
 - Use to fund surveys, PhD students, researchers, *[Eventually]* build platforms
 - Partnership with an organisation really helps

MY PREVIOUS WORK: FOOD BANKS

[RETURN](#)

- Feeding America distributes 100,000 tonnes of food to food banks each year
 - Until 2005, they essentially offered to food banks at random
- The new mechanism has food banks bid on food using tokens
 - Food banks in areas with more poverty get more tokens, so more food
- I looked into the benefit of this new mechanism, and why the mechanism is so popular
 - Big welfare benefits, driven by heterogeneity in what different food banks wanted
 - All about giving food banks more information about the food available (bigger batches)
 - This allowed food banks to “sort” according to the types of food they needed

MY PREVIOUS WORK: COUNCIL HOUSING

[RETURN](#)

-
- Councils face a trade off between “match quality” and “targeting”
 - If give lots of choice, people can wait for a long time (+empty houses)
 - If give little choice, people get houses quickly, but they are unhappy
 - Worked with 6 UK councils investigating these trade offs [horizontal vs vertical heterogeneity]
 - People in TA or HA properties would often be happy to wait for the perfect house
 - Proposed the “Multiple Waiting List” design [Thakral, 2016]
 - People queue for “any” house, and get offered first house available.
 - if they reject the house, they can join a queue for more specific types of house (e.g. geography)
 - Estimated that this could cut waiting times by 40%
 - + reduce the time houses spent vacant by a similar amount
 - *[more recently]* Working to understand scope of demand for council house swaps

MY PREVIOUS WORK: RIDE SHARING

[RETURN](#)

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- Blablacar matches those with a spare seat in their car, with those looking to travel
 - Predominantly inter-city travel, with non-professionals (unlike Uber)
 - Most matches happen <72 hours before departure
 - 50% of Seats unfilled, 60% of passengers without seats
 - Is this a preference? Or is there something wrong with the platform?
 - Currently trialing a mechanism to encourage earlier matches [seat guarantees]
 - Removing additional frictions to forming quick matches
 - Early results:
 - Matches don't occur earlier
 - But more matches occur

MY PREVIOUS WORK: DOCTOR ALLOCATION

[RETURN](#)

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- Junior doctors are allocated to Foundation Programmes (first proper hospital position)
 - Doctors rank programmes. Programmes rank doctors. An algorithm determines the match.
 - Doctor ranking was previously based on exam results
 - In 2023 they changed the algorithm and how doctors are ranked.
 - Moved to a “strategic” algorithm – the higher you rank the programme, the more likely you get in
 - Doctor ranking is now entirely random
 - We are investigating if this will achieve their aims, whether there are better alternatives
 - Early results suggest exam results fell, but no impact on doctor quality
 - It is harder to know how you will place and how you **should** apply
 - We are investigating differential impacts on doctors from different backgrounds
 - + How will this impact retention of junior doctors?

DOCTOR ALLOCATION: CHALLENGES AND LESSONS

FROM ROTH (1986)

[RETURN](#)

- Rural Hospitals
 - No one wants to be in a rural hospital. Rural hospitals end up with lower quality doctors
 - The choice of algorithm doesn't help. You have to make the hospital more attractive somehow
 - This is particularly a problem for allocating teachers to schools
- Couples
 - Many doctors have a partner who is also a doctor
 - This can create *instability*, due to their complementarities
 - Rough solution: match them together, or match them first

DOCTOR ALLOCATION: TWO ALGORITHMS

RETURN

- Immediate Acceptance:

EXAMPLE

- Step 1: Assign as many doctors their first choice hospital as possible.
 - If more doctors list a hospital first than jobs available, keep the most preferred doctors.
- Step N: For any currently unassigned, assign as many doctors their Nth Choice as possible
 - If more doctors list a hospital Nth than jobs available, keep the most preferred doctors
- Terminate once all doctors are assigned

- Deferred Acceptance:

EXAMPLE

- Step 1: Assign as many doctors their first choice hospital as possible.
 - If more doctors list a hospital first than jobs available, keep the most preferred doctors
- Step 2: For any currently unassigned, assign as many doctors their second choice hospital as possible.
 - The hospital keeps their favourite doctors out of all currently assigned (old and new), the rest return to being unassigned
- Step N: For any currently unassigned, assign as many doctors their Nth Choice as possible
 - The hospital keeps their favourite doctors out of all currently assigned (old and new), the rest return to being unassigned
- Terminate once all doctors are assigned

SCHOOL CHOICE: THE BENEFITS OF CENTRALISATION

FROM ABDULKADIROGLU, AGARWAL, AND PATHAK (2017)

[RETURN](#)

- New York Public High Schools went from decentralised to centralised assignment in 2003
 - Students used to receive many offers, or none.

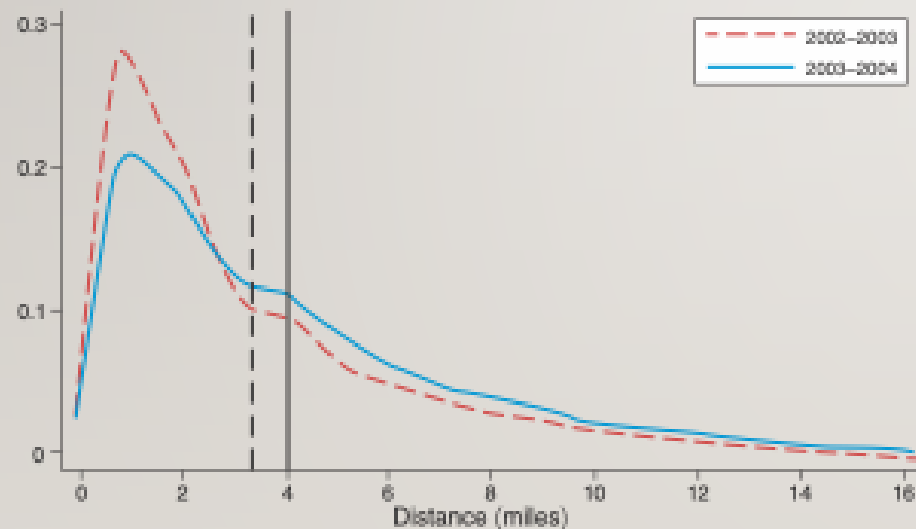


FIGURE 2. DISTRIBUTION OF DISTANCE TO ASSIGNED SCHOOL IN UNCOORDINATED (2002–2003) AND COORDINATED (2003–2004) MECHANISM

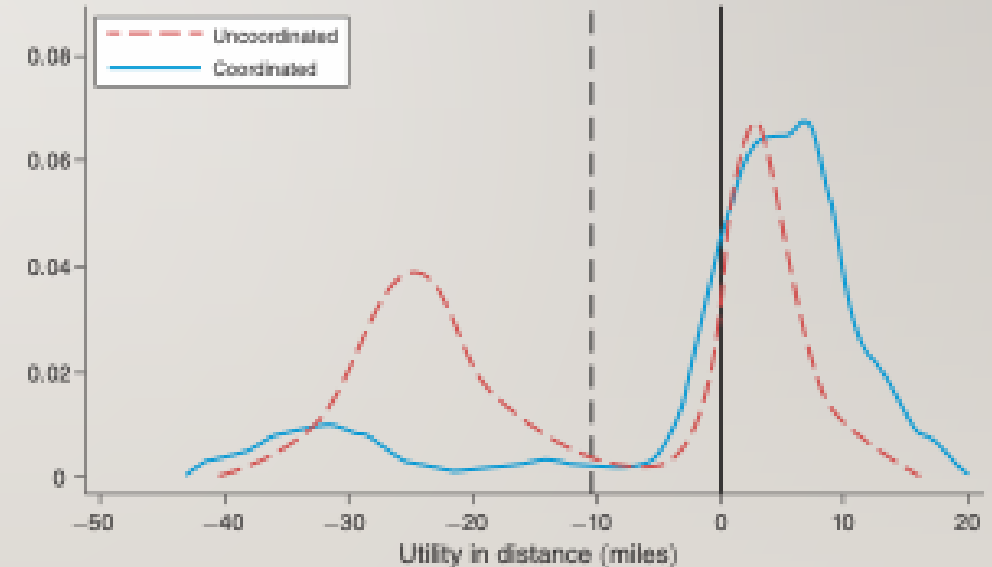


FIGURE 4. STUDENT WELFARE FROM UNCOORDINATED AND COORDINATED MECHANISM

SCHOOL CHOICE: CHOICE OF MECHANISM

FROM AGARWAL AND SOMAINI (2020)

[RETURN](#)

- Many school districts use mechanisms that require strategising:
 - If you list a competitive school in your top choice, you may miss out on your second choice
- This has been found to be harmful
 - It harms students who “incorrectly” strategise, typically low SES students
 - It also harms students who do not know exactly how competitive a school is
- However, it can have benefits, depending on the circumstances:
 - If preferences for schools are not strongly correlated
 - If schools are not so competitive

CADET MATCHING: THE PROBLEM

FROM SONMEZ AND SWITZER (2013)

[RETURN](#)

- The US army allocates newly qualified cadets to their first posting
 - Barracks want the best Cadets, based on a written + physical exam score
 - Cadets are matched to a barracks, a specialisation, and a length of term (3 or 5 years)
- Allocation is (now) based on the same algorithm used for doctors
 - Cadets get a score boost for a particular post if apply for 5 year term
- This score boost has been very effective:
 - This score boost has been found to increase enrolment terms
 - And also boost moral!
 - Cadets with a strong preference for a particular posting use 5 year term as a **signal**

[PREVIOUSLY](#)



CADET MATCHING

THE OLD ALGORITHM

RETURN

- They previously used an extremely complicated algorithm
 - (though the score boost was still in place and did seem to boost enrolment still)
- Cadets reported discontent with this algorithm
 - It was unclear how they would place, and how effort lead to improved outcomes
 - They also felt *justified envy*: lower scoring candidates getting the positions they wanted
- They switched to the same algorithm used to allocate doctors
 - It incentivises truth telling (transparency)
 - And it turns out it prevents justified envy

TEACHER ALLOCATION: OTHER CONCERNS

[RETURN](#)

- Distribution of Teachers
 - A very active area of research, e.g. diversity requirements
 - Allow swaps only if they improve diversity [CDTTU, (2025)]
- Value added concerns
- Retention / private schools

TEACHER ALLOCATION: TOP TRADING CYCLES

FROM SHAPLEY AND SCARF (1974)

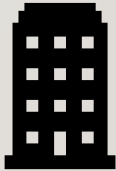
[RETURN](#)

- The Top Trading Cycles Algorithm
 - Also, used by Hermit Crabs for thousands of years: <https://www.youtube.com/watch?v=fldnocPQXDQ>
- Every Teacher points at their favorite School – *this can be their current school*
 - Schools also “point” (metaphorically) at their assigned teacher
- Find a chain:
 - Theorem: there will always be at least one chain – e.g. all the teachers pointing at their own school
 - Finding chains is a linear algebra problem: With N teachers, this involves at most N^2 calculations
- Execute a chain:
 - The teachers all get the school they are pointing at, and are removed from the problem [MATCHED]
 - Theorem: there may be multiple chains, but it doesn't matter* which you choose! (the chain will be there next round)
- Repeat until every teacher is MATCHED. This can be shown to take around $N \log(N)$ calculations (i.e. $< N^2$)
 - The algorithm can also be easily adapted to incorporate new teachers, and vacancies
 - It can also incorporate constraints [e.g. teacher qualifications], teacher scores, and other school preferences

[EXAMPLE](#)

DEFERRED ACCEPTANCE

RETURN

Ahmed	$\{1, 0\}$			$\{C, A, D, B, 0\}$	1
Beth	$\{1, 3, 2, 4, 0\}$			$\{A, D, 0\}$	2
Claire	$\{2, 3, 4, 1, 0\}$			$\{B, D, C, 0\}$	3
David	$\{3, 4, 1, 0\}$			$\{B, C, D, A, 0\}$	4

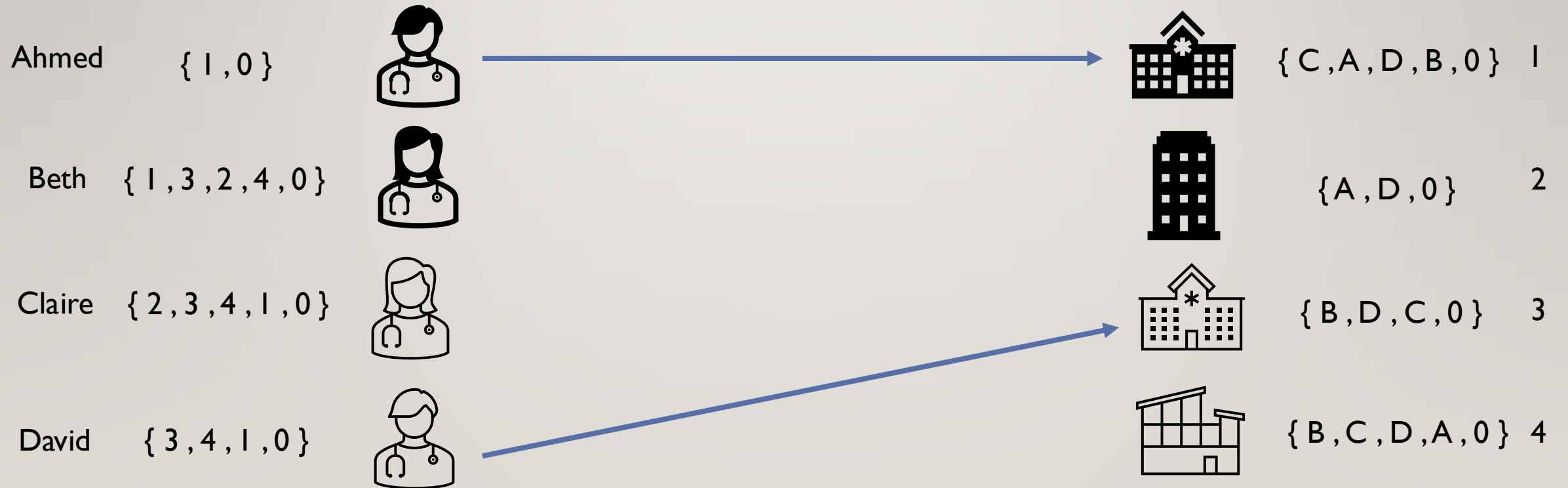
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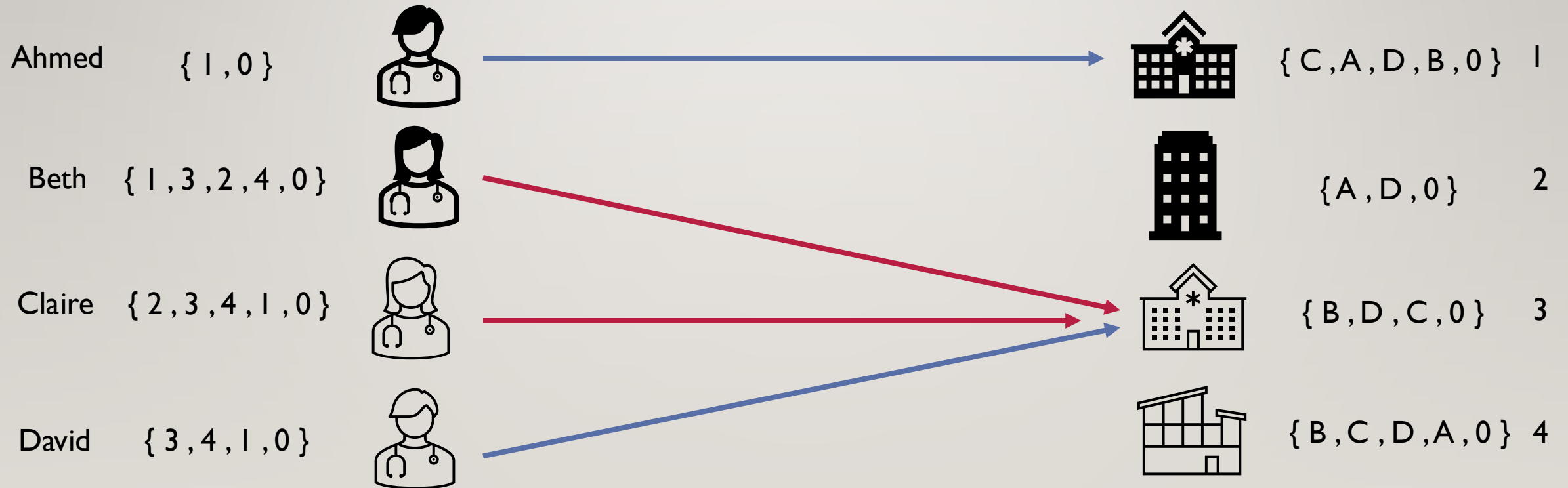
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DEFERRED ACCEPTANCE

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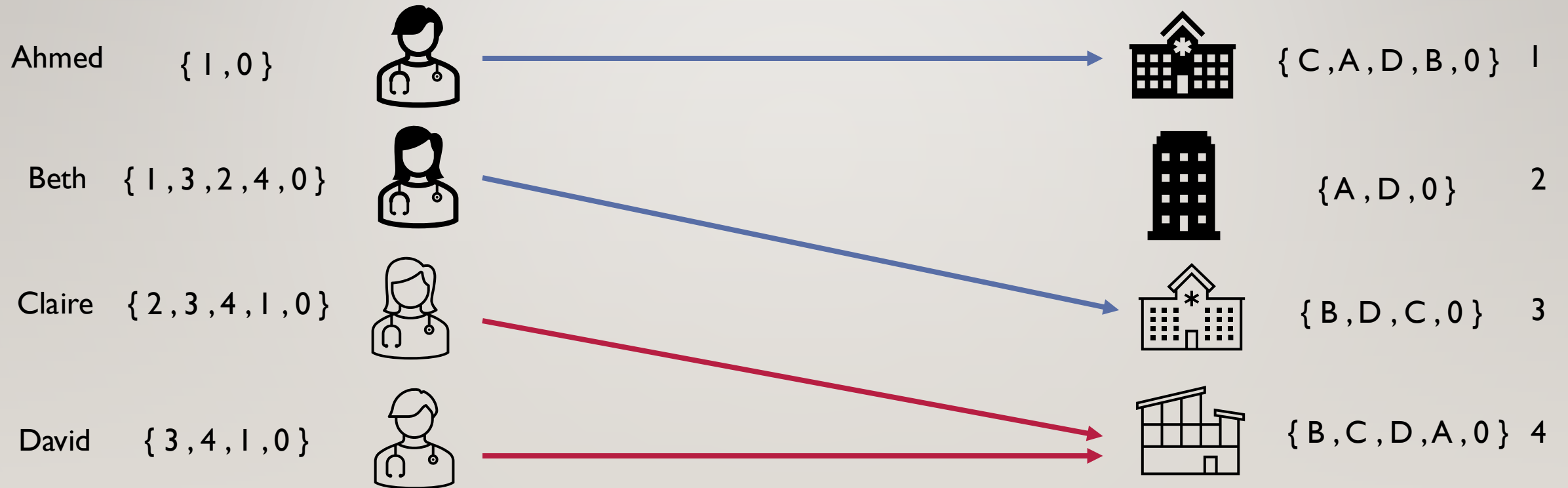
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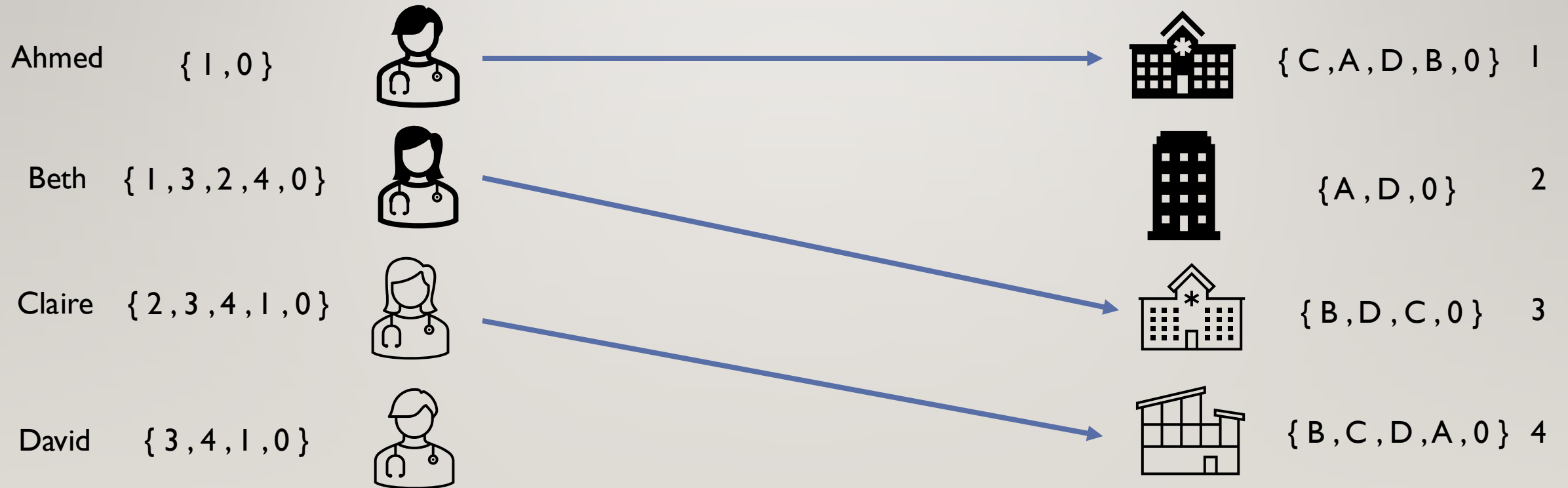
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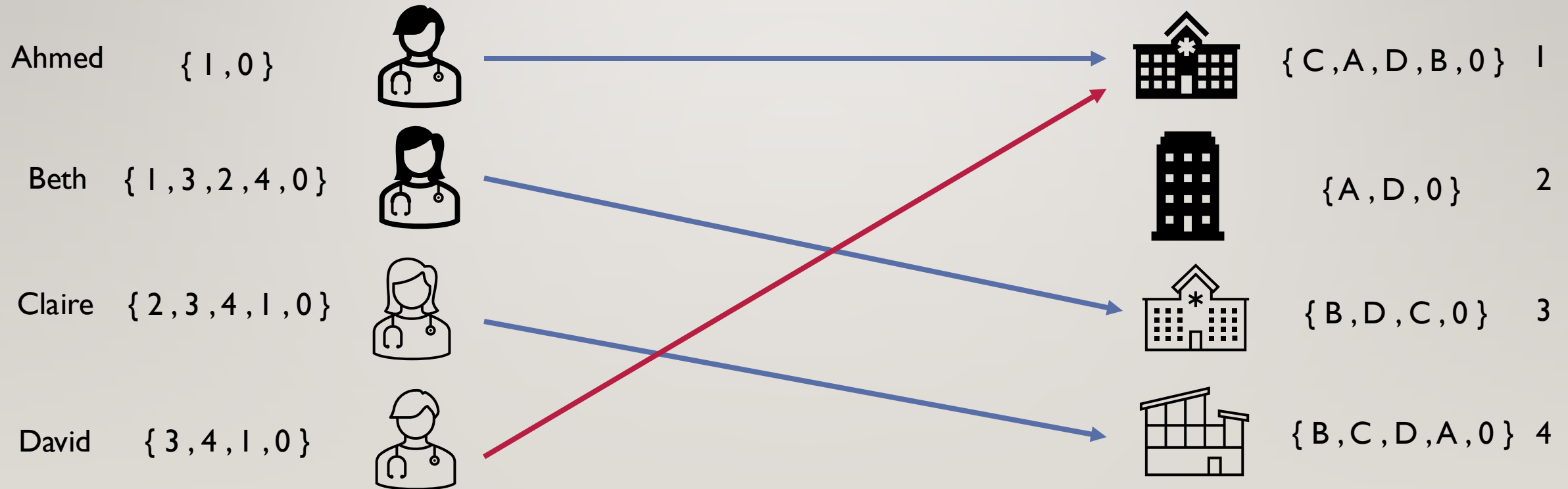
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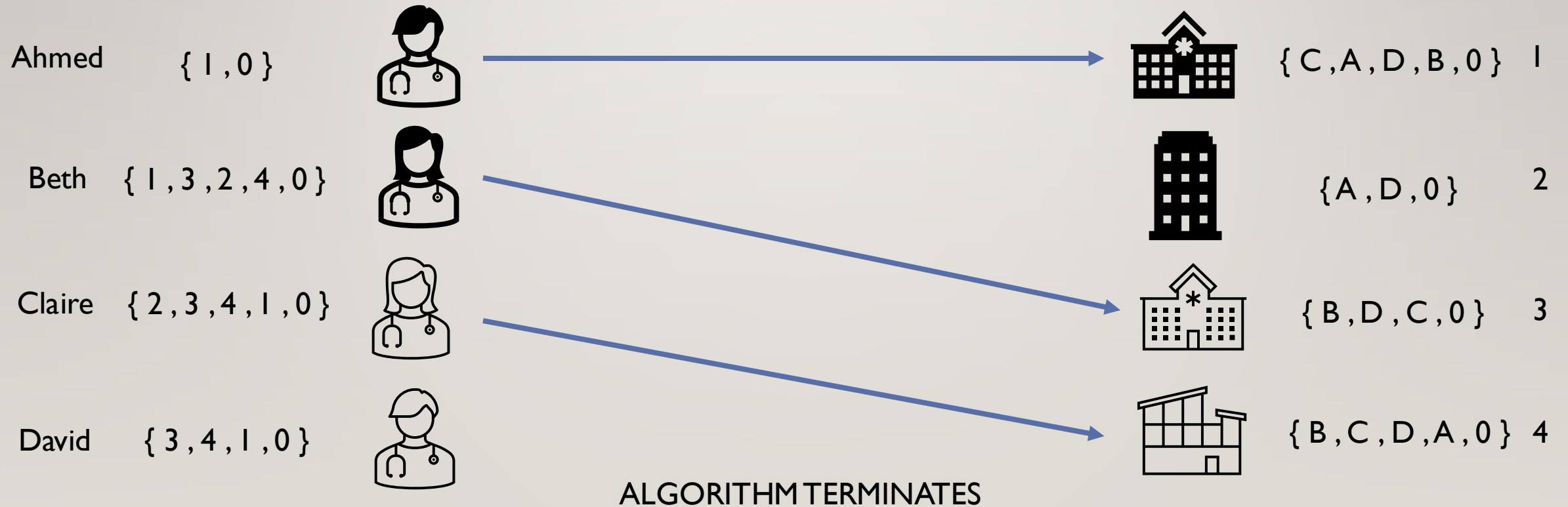
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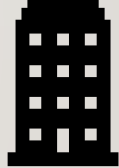
DEFERRED ACCEPTANCE

RETURN



IMMEDIATE ACCEPTANCE

RETURN

Ahmed	{ 1 , 0 }			{ C , A , D , B , 0 }	1
Beth	{ 1 , 3 , 2 , 4 , 0 }			{ A , D , 0 }	2
Claire	{ 2 , 3 , 4 , 1 , 0 }			{ B , D , C , 0 }	3
David	{ 3 , 4 , 1 , 0 }			{ B , C , D , A , 0 }	4

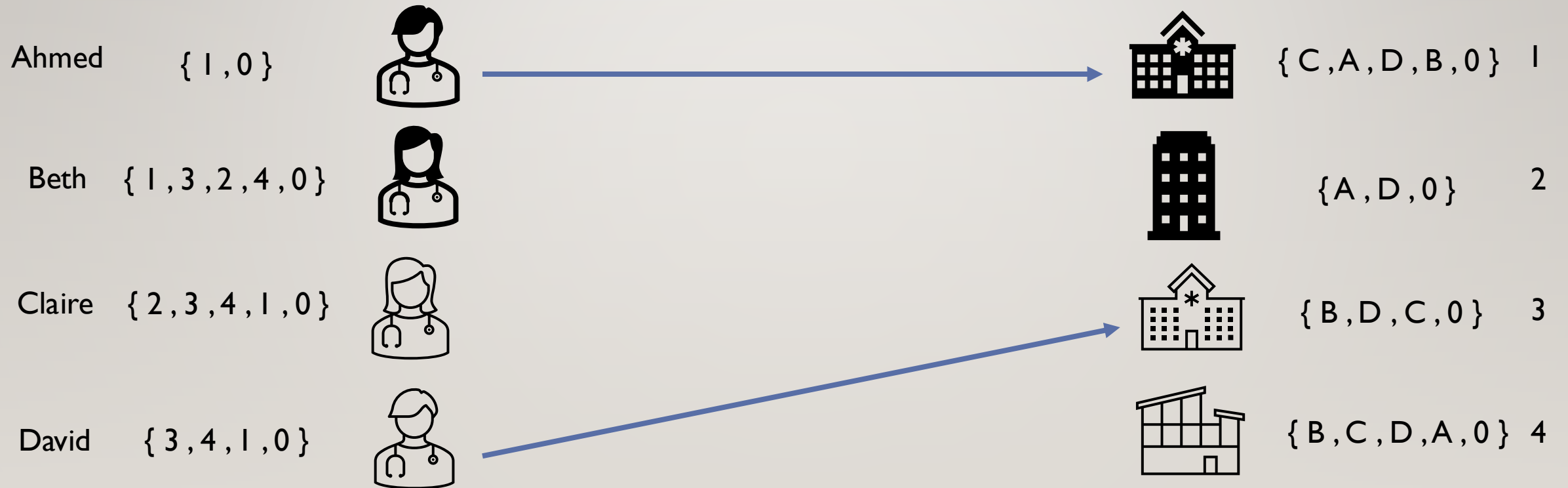
IMMEDIATE ACCEPTANCE

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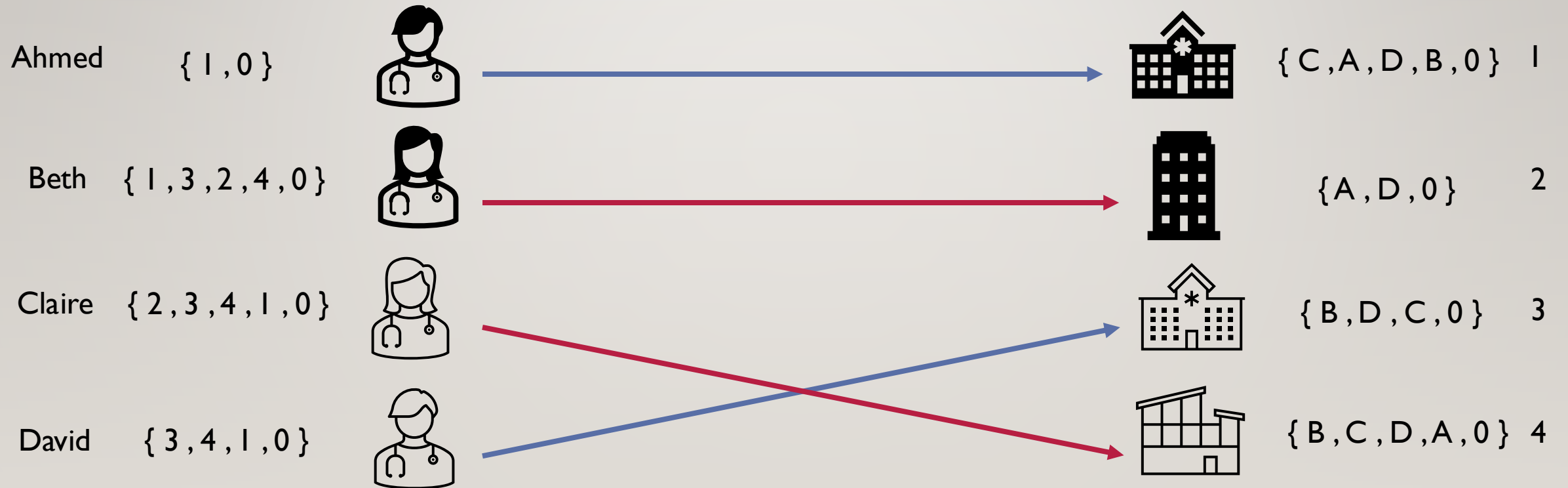
IMMEDIATE ACCEPTANCE

RETURN



IMMEDIATE ACCEPTANCE

RETURN



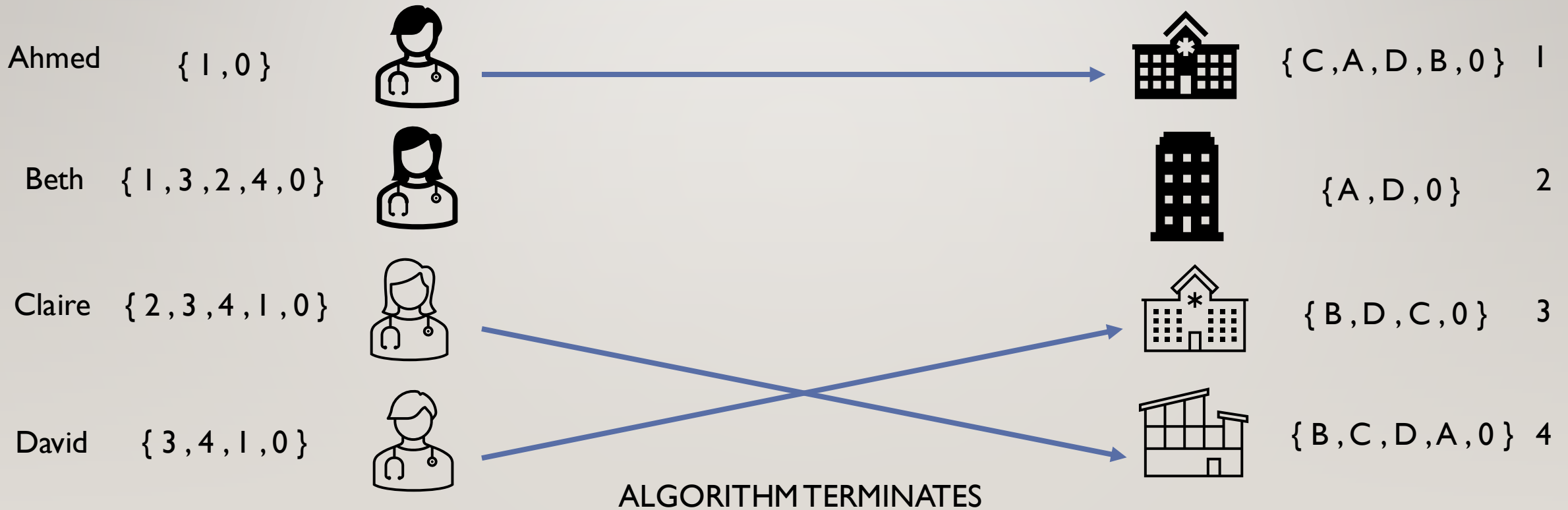
IMMEDIATE ACCEPTANCE

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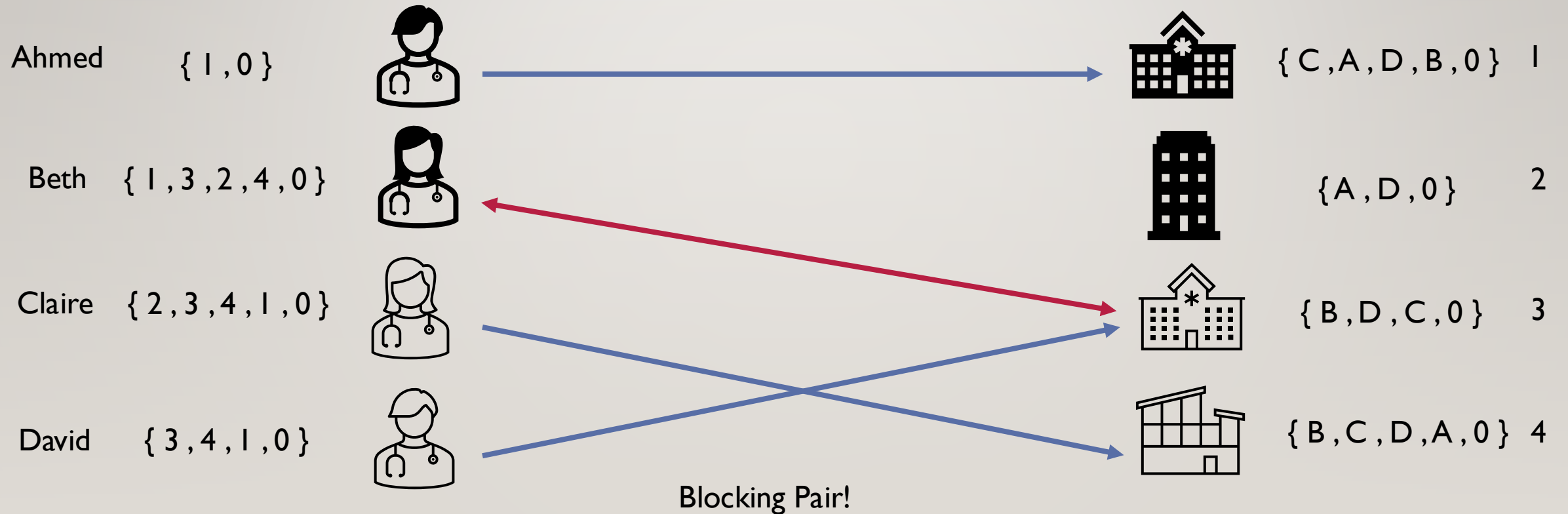
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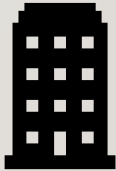
IMMEDIATE ACCEPTANCE

RETURN



IMMEDIATE ACCEPTANCE

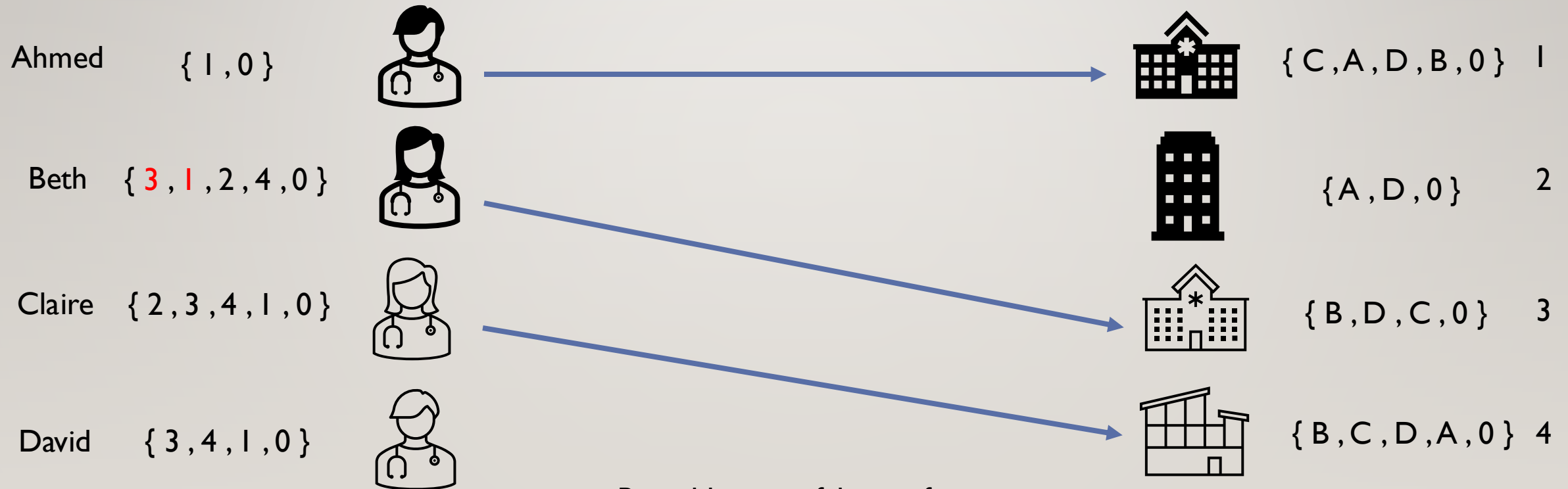
RETURN

Ahmed	$\{1, 0\}$			$\{C, A, D, B, 0\}$	1
Beth	$\{3, 1, 2, 4, 0\}$			$\{A, D, 0\}$	2
Claire	$\{2, 3, 4, 1, 0\}$			$\{B, D, C, 0\}$	3
David	$\{3, 4, 1, 0\}$			$\{B, C, D, A, 0\}$	4

B could report false preferences

IMMEDIATE ACCEPTANCE

RETURN



B could report false preferences

TOP TRADING CYCLES

RETURN

Ahmed { 3 , 1 , 4 , 2 , 0 } 

Beth { 1 , 4 , 0 } 

Claire { 2 , 4 , 3 , 0 } 

David { 2 , 3 , 4 , 1 , 0 } 



{ A , B , C , D , 0 } 1



{ C , A , B , D , 0 } 2



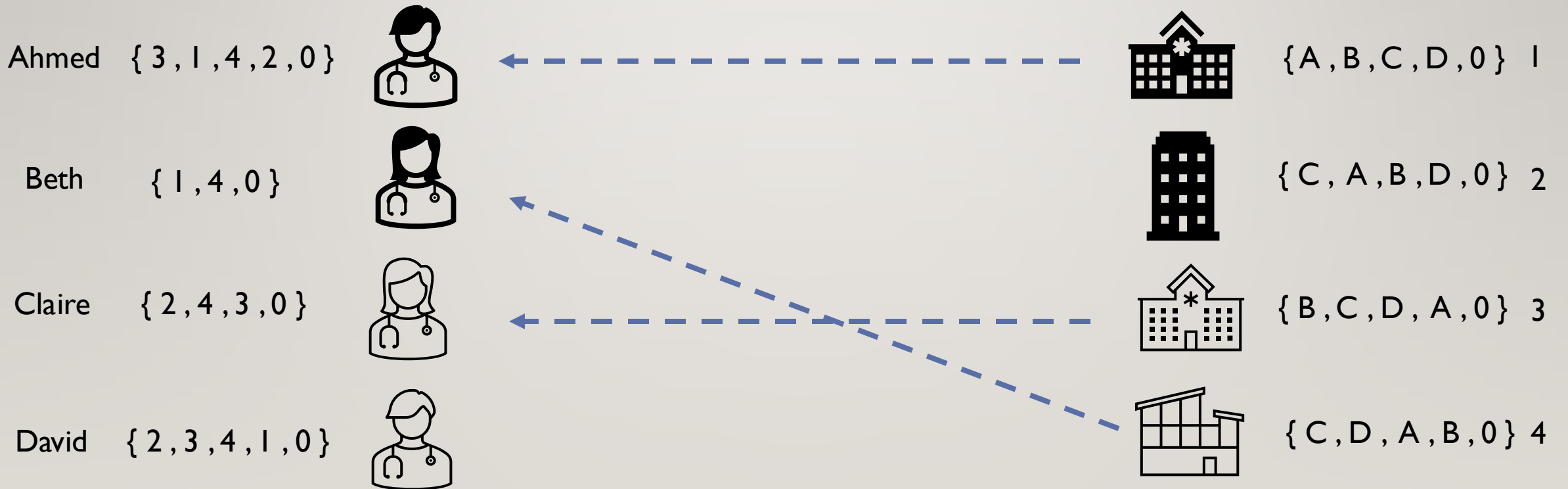
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{ C , D , A , B , 0 } 4

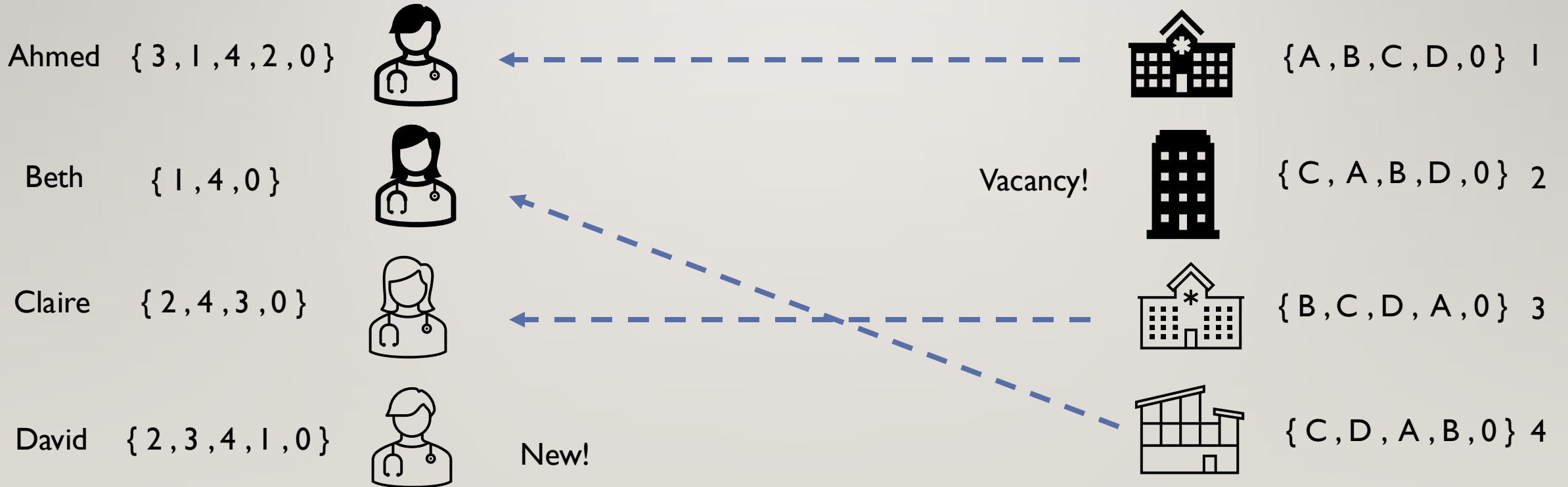
TOP TRADING CYCLES

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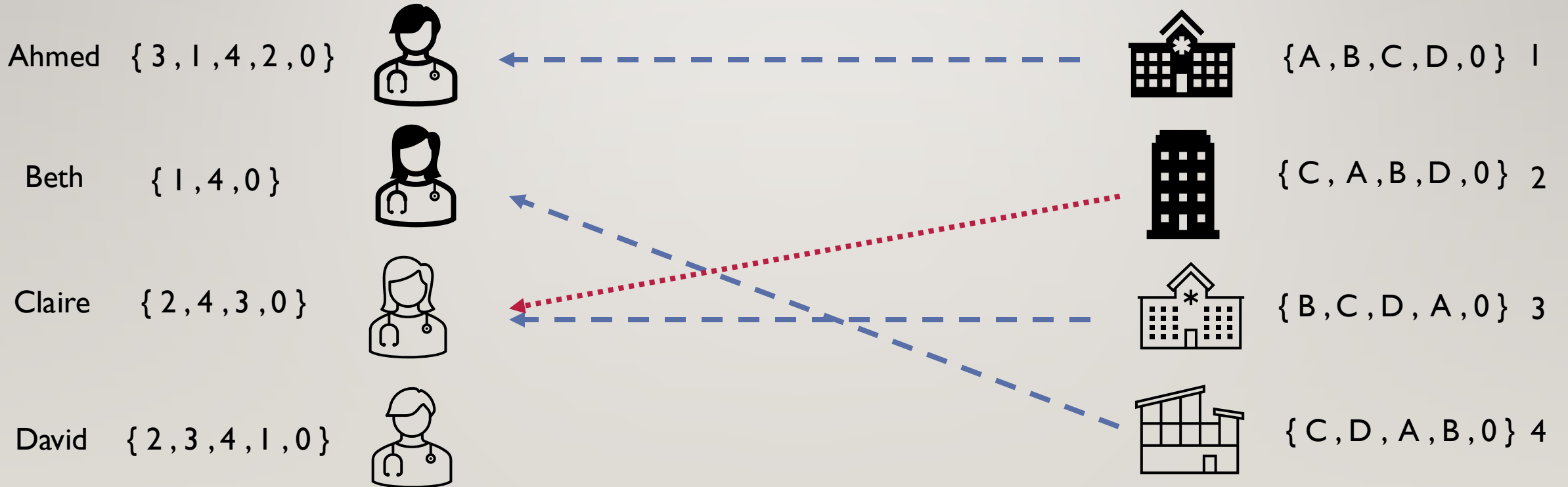
TOP TRADING CYCLES

RETURN



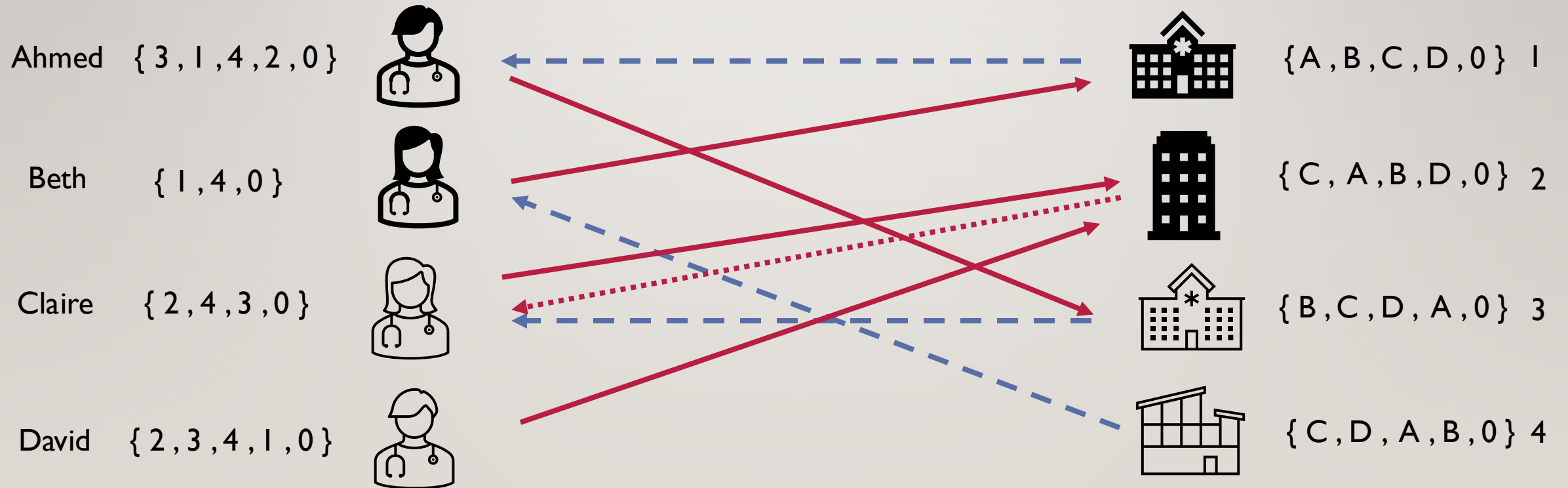
TOP TRADING CYCLES

RETURN



TOP TRADING CYCLES

RETURN



TOP TRADING CYCLES

RETURN



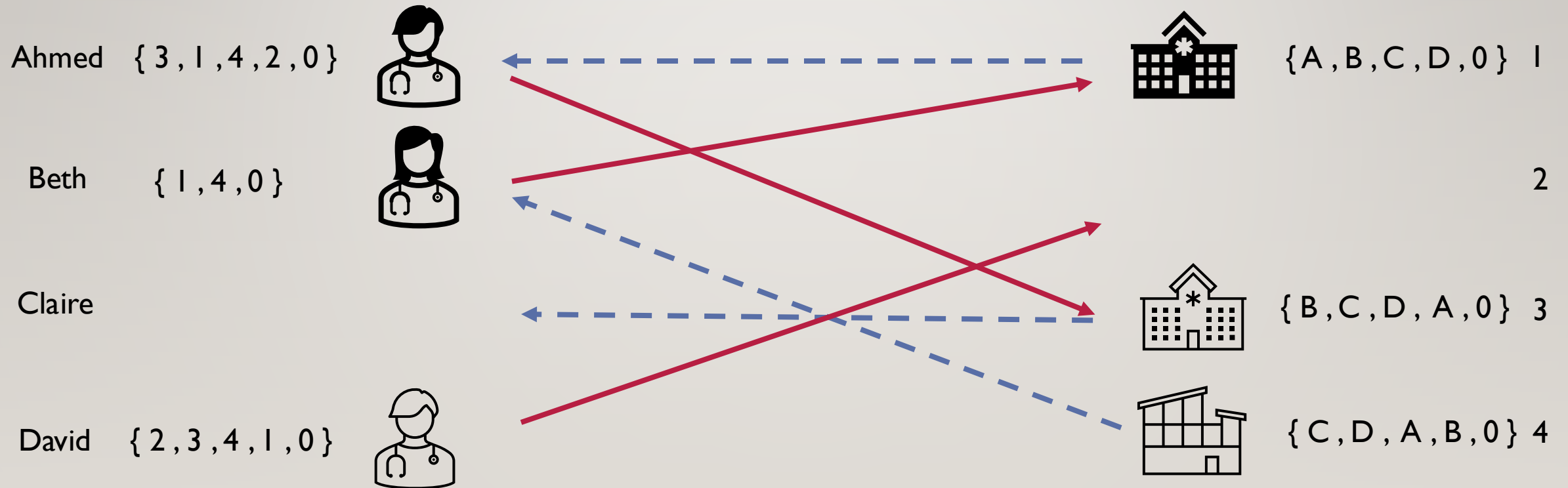
TOP TRADING CYCLES

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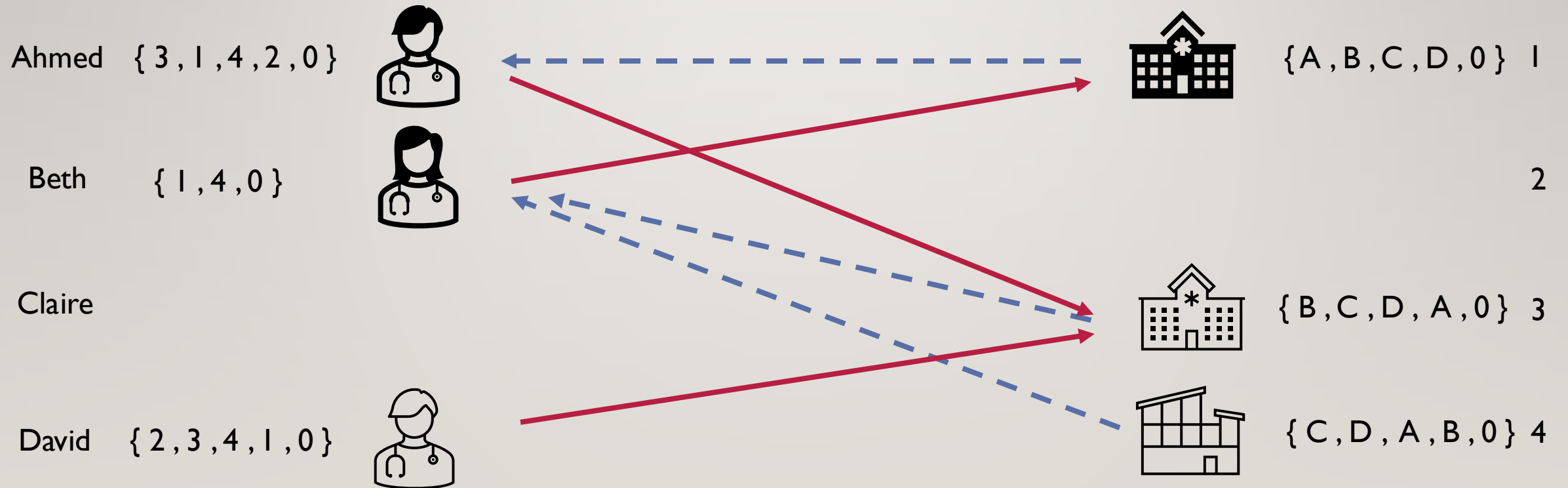
TOP TRADING CYCLES

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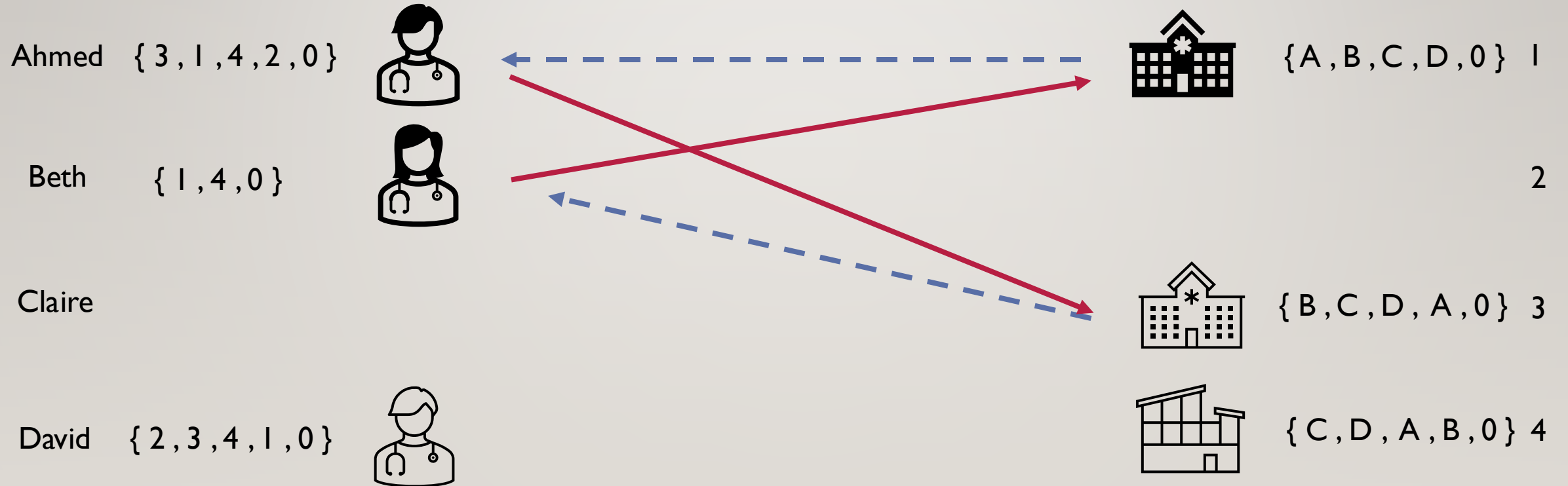
TOP TRADING CYCLES

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TOP TRADING CYCLES

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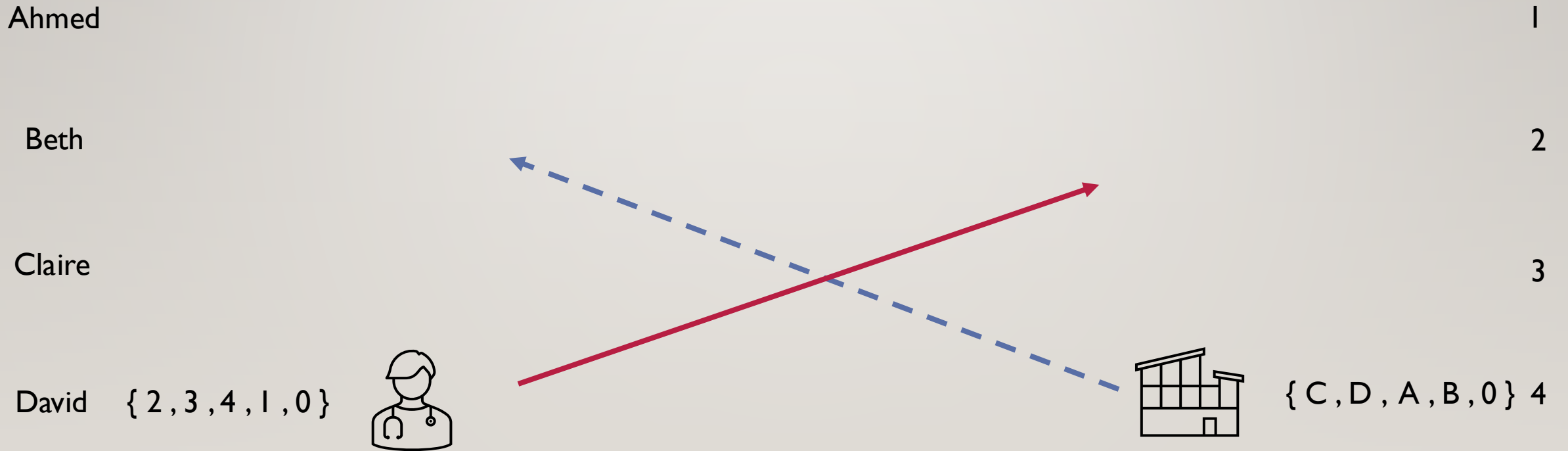
TOP TRADING CYCLES

RETURN



TOP TRADING CYCLES

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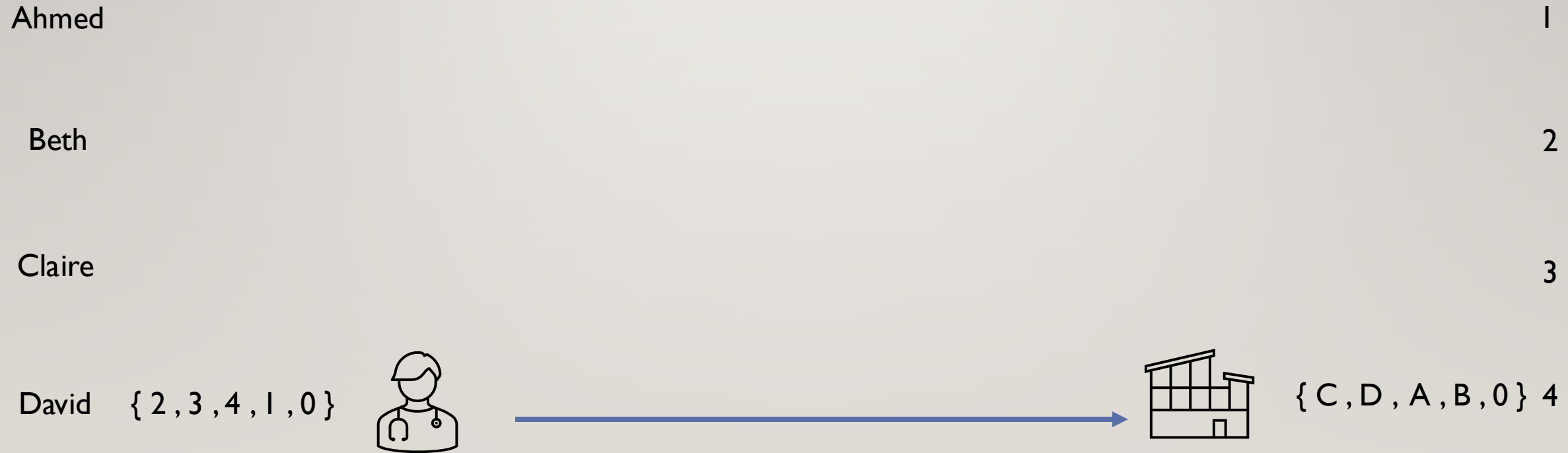
TOP TRADING CYCLES

RETURN



TOP TRADING CYCLES

RETURN



TOP TRADING CYCLES

RETURN

Ahmed	1
Beth	2
Claire	3
David	4

ALGORITHM TERMINATES

TOP TRADING CYCLES

RETURN

